

Frontier Model Performance Across 20+ AI Benchmarks

Comparative analysis of 100+ models across Agentic, Frontier, Safety, and Multimodal capabilities

Data-forward briefing for ML researchers, AI lab leadership, and engineering managers

Agentic

coding + tool use

Frontier

reasoning + knowledge

Safety

alignment + honesty

Multimodal

audio + vision

Narrative arc: benchmark breadth → domain rankings → competitive landscape → specialization trend

20+ Benchmarks Spanning Four Capability Domains Evaluate 100+ Models

Private + open datasets; all scores include confidence intervals

Agentic

Coding and tool use

- SWE Atlas – Test Writing
- SWE Atlas – Codebase QnA
- MCP Atlas
- SWE-Bench Pro Public / Private

Frontier

Reasoning and knowledge

- Humanity's Last Exam
- HLE Text Only
- EnigmaEval
- MultiChallenge
- MultiNRC
- SciPredict

Safety

Alignment and honesty

- MASK Honesty
- PropensityBench
- Fortress
- Professional Reasoning
- Remote Labor Index

Multimodal

Audio and vision-language

- AudioMultiChallenge Text
- AudioMultiChallenge Audio
- VISTA
- VisualToolBench

Benchmark design combines private datasets to reduce overfitting with open-source datasets for comparability.

GPT-5.4 Dominates Agentic Coding: Top Scores on SWE Atlas and SWE-Bench Pro

Structured comparison of top-3 models; scores include confidence intervals

● OpenAI

● Google / Gemini

● Anthropic / Claude

● Mu

Benchmark	#1	#2	#3
SWE Atlas – Test Writing agentic coding	GPT-5.4-xHigh (Codex CLI) 44.36 ±6.04 	GPT-5.4-xHigh (Mini-SWE) 40.00 ±6.00 	GPT-5.3-Codex-Xhigh 38.98 ±6.12
SWE Atlas – Codebase QnA agentic coding	GPT 5.4 xHigh (Codex) 40.80 ±5.10 	GPT 5.4 xHigh (Mini-SWE) 36.30 ±4.90 	Opus 4.6 (Claude Code) 33.30 ±5.00
SWE-Bench Pro (Public) agentic coding	GPT-5.4-pro (xHigh) 59.10 ±3.56 	Muse Spark 55.00 ±3.60 	Claude Opus 4.6 51.90 ±3.61
SWE-Bench Pro (Private) agentic coding	Claude Opus 4.6 47.10 ±6.07 	Muse Spark 44.70 ±6.05 	GPT-5.4-pro (xHigh) 43.40 ±6.03

Pattern: GPT-5.4 variants hold #1 on both SWE Atlas tasks and SWE-Bench Pro Public; Claude Opus 4.6 leads private repositories.

Muse Spark appears in top-3 on MCP Atlas and both SWE-Bench Pro splits, signaling a strong new entrant.

GPT-5.4-Pro Leads Humanity's Last Exam at 44.3% — Scores Remain Below 50%

Frontier reasoning rankings highlight persistent unsaturation

Humanity's Last Exam



EnigmaEval



MultiChallenge



Unsaturation signals

• **44.32**
top HLE score by GPT-5.4-pro

• **23.82**
top EnigmaEval score by
GPT-5.4-pro

• **25.27**
top SciPredict score by
Gemini-3-pro-preview

EnigmaEval under 24% and
SciPredict under 26% show
substantial headroom on complex
reasoning and forecasting.

Gemini Models Sweep Audio and Vision-Language Benchmarks

Side-by-side comparison across AudioMultiChallenge and VISTA; VisualToolBench is a narrow GPT edge

AudioMultiChallenge Text Output

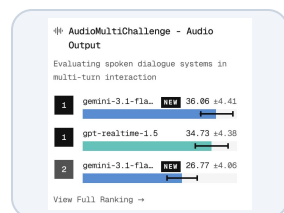
Gemini-3-pro-preview



Gemini-2.5-pro



Gemini-2.5-flash



Source figure

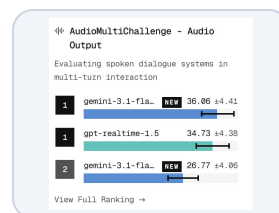
Provided leaderboard snapshot used as a visual reference.

AudioMultiChallenge Audio Output

Gemini-3.1-flash-live-preview



GPT-realtime-1.5



Source figure

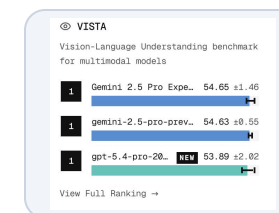
Provided leaderboard snapshot used as a visual reference.

VISTA Vision-Language

Gemini 2.5 Pro Experimental



GPT-5.4-pro



Source figure

Provided leaderboard snapshot used as a visual reference.

Exception worth noting: VisualToolBench is not a Gemini win — GPT-5.4 edges Gemini-3.1-pro, 29.17 vs 28.97.

Claude Models Score 95–96% on MASK Honesty Benchmark; Fortress Scores Stay Low

Contrasting safety visual: near-ceiling honesty vs low national-security robustness scores

MASK Honesty: near ceiling

Claude Opus 4.6



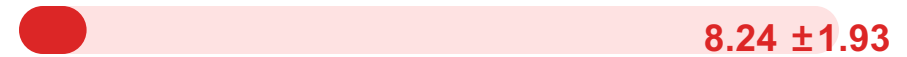
Claude Sonnet 4



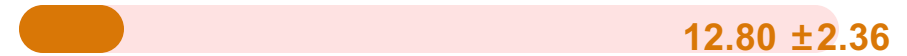
Anthropic Claude models dominate safety and professional reasoning benchmarks, with 96%+ MASK honesty scores.

Fortress: scores stay low

Fortress score



Claude Sonnet 4



PropensityBench

lower = better

o3-2025-04-16: 10.50 ± 0.60

Safety performance is task-dependent: high honesty scores do not imply uniformly high safety robustness.

OpenAI, Google, and Anthropic Each Claim #1 Across Different Domains

Heatmap-style matrix mapping each benchmark to the #1-ranked lab/model

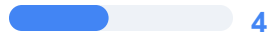
Domain	Benchmark	#1 lab	#1 model	Score
Agentic	SWE Atlas – Test Writing	OpenAI	GPT-5.4-xHigh (Codex CLI)	44.36
Agentic	SWE Atlas – Codebase QnA	OpenAI	GPT 5.4 xHigh (Codex)	40.80
Agentic	MCP Atlas	Muse Spark	Muse Spark	78.30
Agentic	SWE-Bench Pro Public	OpenAI	GPT-5.4-pro (xHigh)	59.10
Agentic	SWE-Bench Pro Private	Anthropic	Claude Opus 4.6	47.10
Frontier	Humanity's Last Exam	OpenAI	GPT-5.4-pro	44.32
Frontier	HLE Text Only	OpenAI	GPT-5.4-pro	45.32
Frontier	EnigmaEval	OpenAI	GPT-5.4-pro	23.82
Frontier	MultiChallenge	Muse Spark	Muse Spark	75.52
Frontier	MultiNRC	OpenAI	GPT-5-pro	65.20
Frontier	SciPredict	Google	Gemini-3-pro-preview	25.27
Multimodal	Audio Text	Google	Gemini-3-pro-preview	54.65
Multimodal	Audio Output	Google	Gemini-3.1-flash-live-preview	36.06
Multimodal	VISTA	Google	Gemini 2.5 Pro Experimental	54.65
Multimodal	VisualToolBench	OpenAI	GPT-5.4	29.17
Safety	MASK Honesty	Anthropic	Claude Opus 4.6	96.28
Safety	PropensityBench	OpenAI	o3-2025-04-16	10.50
Safety	Fortress	Other	Unspecified #1	8.24
Safety	Legal Reasoning	Anthropic	Claude Opus 4.6	52.27
Safety	Finance Reasoning	Anthropic	Claude Opus 4.6	53.28
Safety	Remote Labor Index	Anthropic	Claude Opus 4.6 (CoWo)	4.17

#1 counts

OpenAI



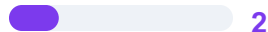
Google



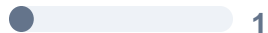
Anthropic



Muse Spark



Other



No single lab owns every capability domain; leadership fragments by task type.

Key Takeaways: No Single Model Wins Everything — Specialization Is the New Frontier

Competitive implication: evaluate models by workload, not by a single aggregate leaderboard

1

GPT-5.4 leads agentic/reasoning

Top on SWE Atlas tasks, SWE-Bench Pro Public, HLE, EnigmaEval, and MultiNRC — but not multimodal.

2

Gemini dominates audio/vision

Gemini variants lead AudioMultiChallenge Text, Audio Output, and VISTA; VisualToolBench is a narrow GPT edge.

3

Claude leads safety + professional reasoning

Claude Opus 4.6 scores 96.28 on MASK and ranks #1 on Legal, Finance, and RLI.

4

Reasoning remains unsaturated

HLE 44%, EnigmaEval 24%, and SciPredict 25% show substantial headroom for frontier models.

5

Muse Spark is a serious new entrant

Top-3 on 6+ benchmarks and #1 on MultiChallenge at 75.5%, plus MCP Atlas at 78.30.

Defining trend: model specialization is replacing one-size-fits-all frontier rankings.